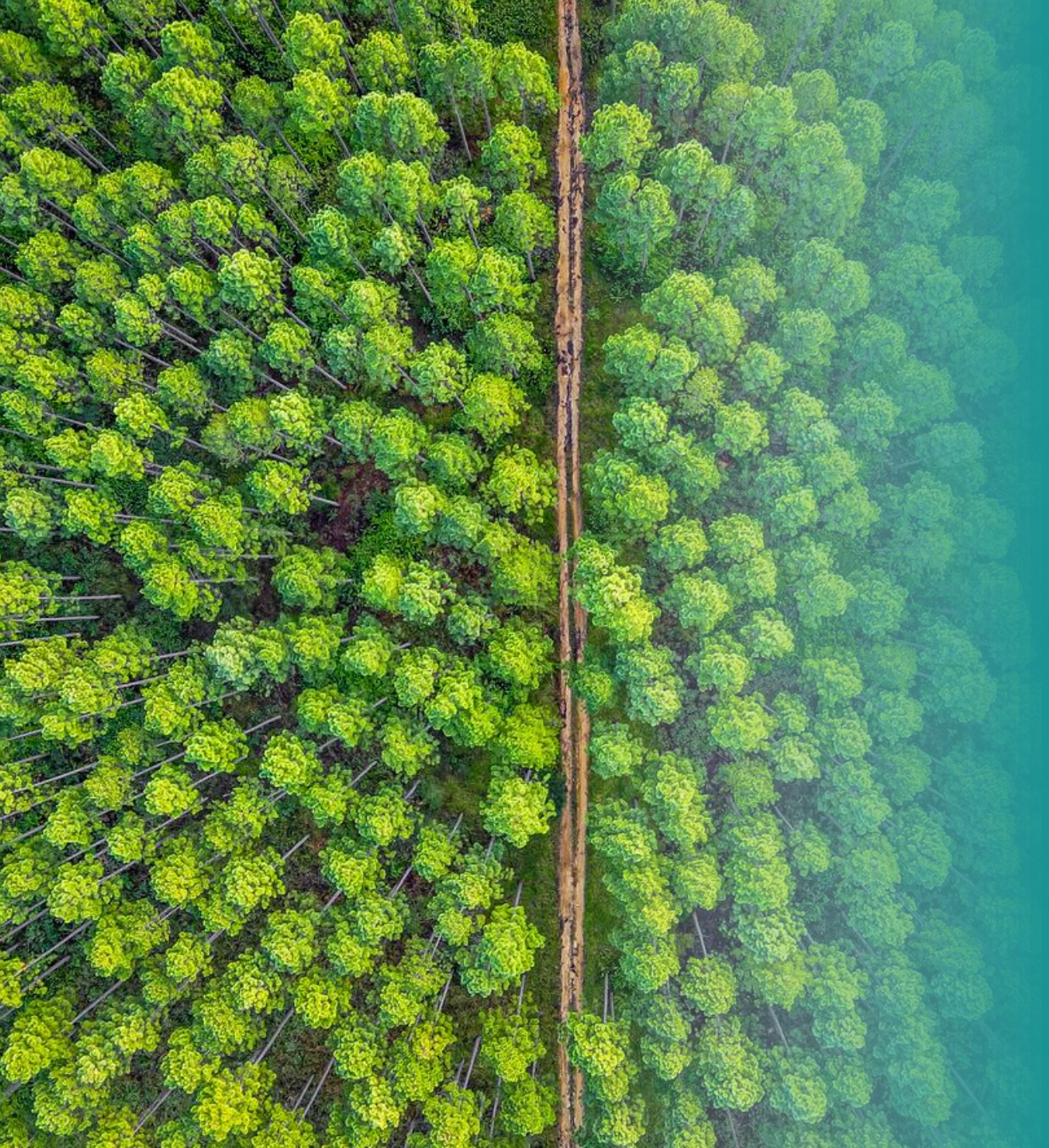


“Causal Inference and Counterfactuals in Earth Observation Research”

Geo4Impact 2025

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May 28, 2025

Situating the chapter in the book | Clarifying our key motivations

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CHAPTER MOTIVATION:

- Introduce non-economists to terms, concepts, tools, and objectives of causal inference/IEs
- Encourage applied economists to incorporate EO into their methods
- Provide a "stratospheric" view of key IE methods, enough to get people started
- Unify elements of GIE that would otherwise be repeated N-fold across multiple methods chapters
- Instill the recognition that the appropriateness of an IE method should be right-sized to the specific research question and the prevailing empirical context

Learning objectives | Readers of this chapter should be able to …

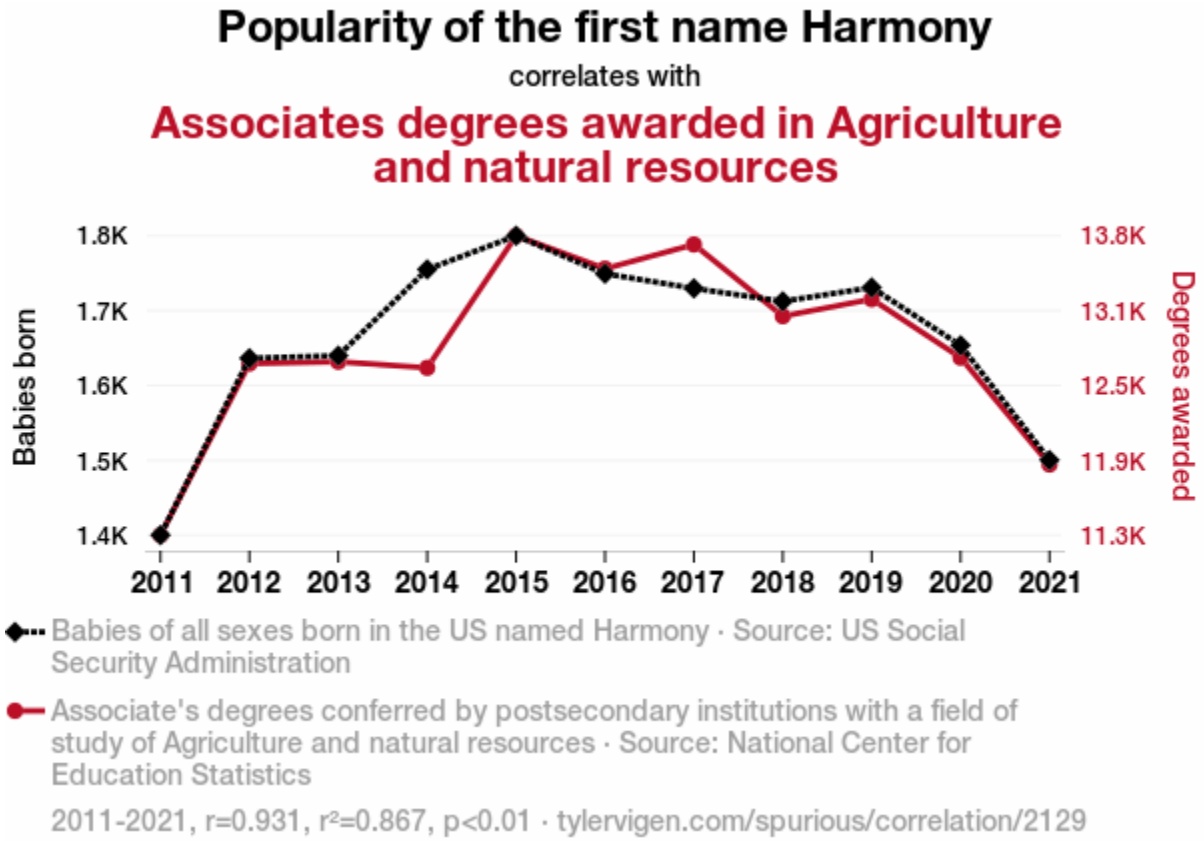
- Determine the most suitable causal inference methodology for a given implementation context, including the timing and nature of treatment assignment and which outcomes remote sensing can provide information about
- Identify key considerations a researcher should address in using remotely sensed data to carry out a geospatial impact evaluation
- Explain how remote sensing data can both broaden and deepen the scope of research questions that impact evaluations can answer

High-level organization of chapter | Summarizing semester course into a chapter

- Very brief history of causal research in social sciences
- Clarifying definitions and key terms from causal inference methods
- Common IE methods for GIEs
  - **Experimental**
    Randomized controlled trials
  - **Quasi-experimental**
    Matching methods
    Difference in differences
    Regression discontinuity → spatial regression discontinuity
    Synthetic control method
- Best practices in improving the use of EO to support causal identification
- Conclusion and exercise questions

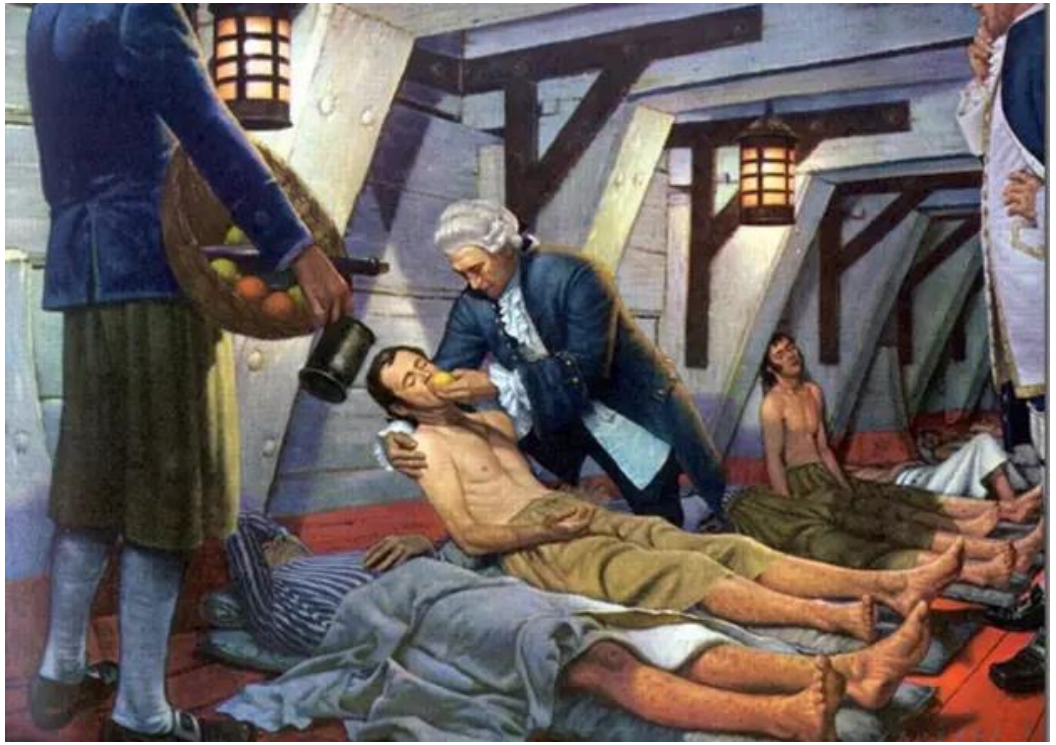
Very brief history of causality | Scurvy, cholera, and whooping cough

Annoying but true: correlation is not causation.



This fortunate scurvy-afflicted sailor was assigned to the “citrus” (2 oranges + 1 lemon) treatment arm; others were administered less effective supplements:

- Cider, elixir of vitriol, sea-water, garlic and horse-radish paste, vinegar



James Lind - 1747

https://www.bbc.com/news/uk-england-37320399

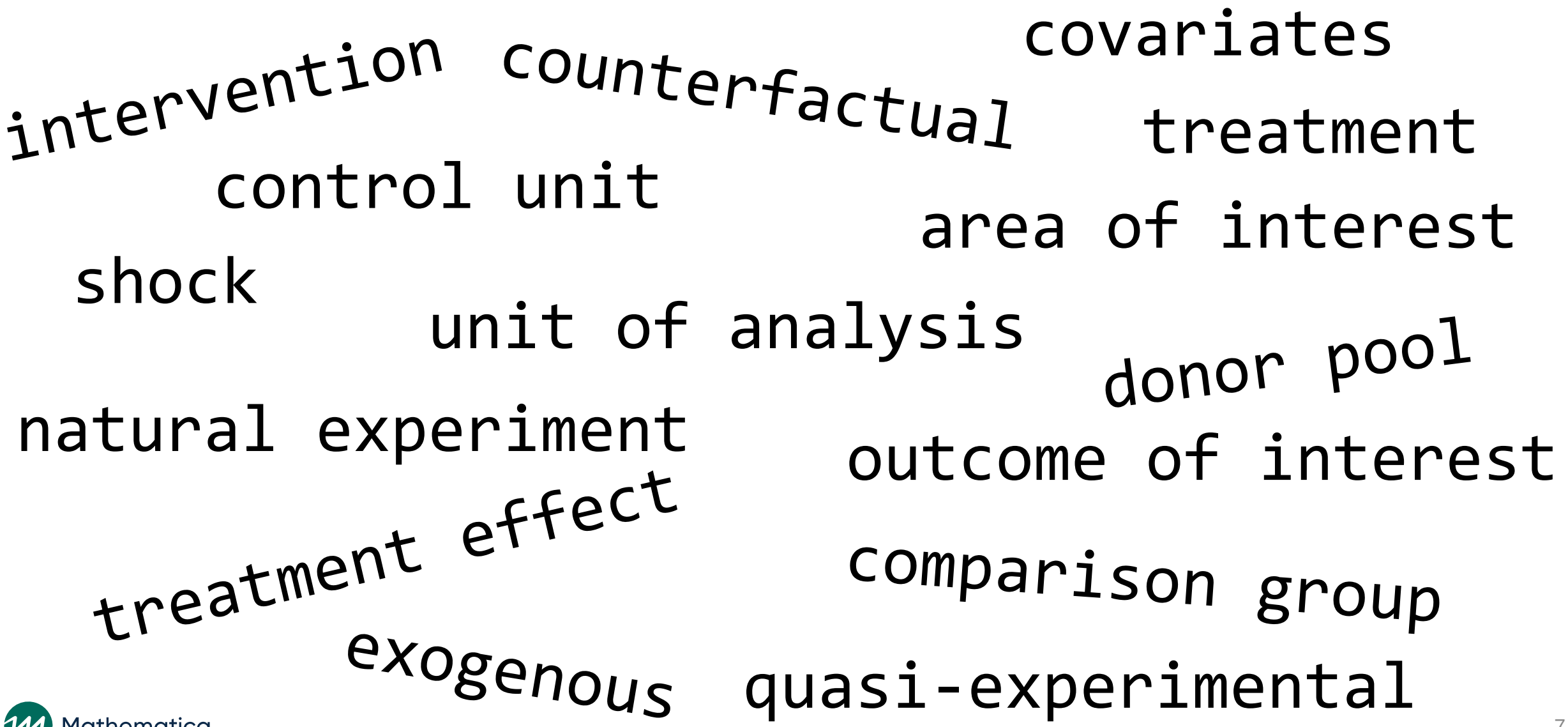
Fundamental problem of CI | Cannot observe both T + C states for an individual unit

- Motivate the centrality of **average treatment effect** estimation, not individual treatment effect
- Q: How does intercropping affect productivity on *my* plot?
- Sets stage for explaining **counterfactuals** and what might constitute a defensible counterfactual





Learning a foreign language | Explaining foundational concepts



Anatomy of a GIE | Multiple avenues for integrating EO into a GIE

$$y_{it} = \beta Treat_i + \boldsymbol{X}'_{\boldsymbol{it}}\delta + \epsilon_{it}$$

EO data Non-EO data

Change in forest canopy cover over time	$y_{it} = \beta Treat_i + \boldsymbol{X}'_{\boldsymbol{it}}\delta + \epsilon_{it}$	
Vegetation cover	$y_{it} = \beta Treat_i + \boldsymbol{X}'_{\boldsymbol{it}}\delta + \epsilon_{it}$	Fire exposure
Vegetation cover	$y_{it} = \beta Treat_i + \boldsymbol{X}'_{\boldsymbol{it}}\delta + \epsilon_{it}$	Fire exposure and proximity to roads
	$y_{it} = \beta Treat_i + \boldsymbol{X}'_{\boldsymbol{it}}\delta + \epsilon_{it}$	Hurricane exposure and slope
	$y_{it} = \beta Treat_i + \boldsymbol{X}'_{\boldsymbol{it}}\delta + \epsilon_{it}$	Plot size at baseline (RD and land redistribution scheme)

Survey or administrative data

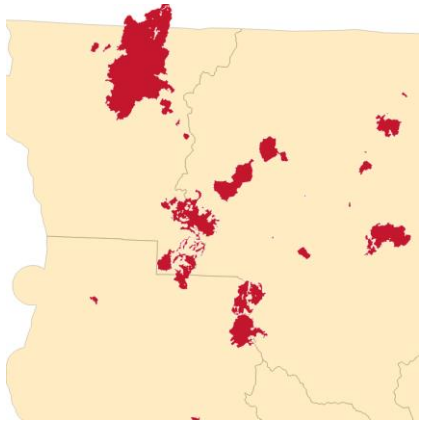
Some strengths of EO data | Supplementing and substituting ‘traditional’ data

Finer accuracy over treatment area

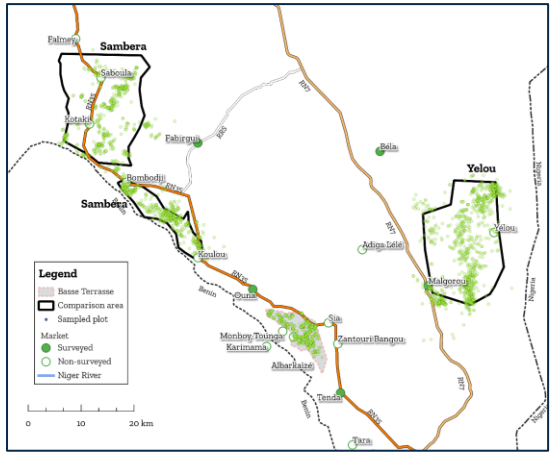
~Zero marginal cost of extending geographic coverage

Continuous monitoring of challenging environments

Temporal coverage



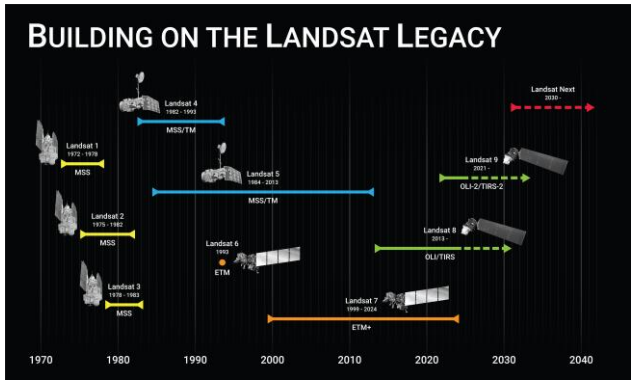
California wildfires (2023, 2024) and county borders



Ksoll et al. 2023 – MCC Niger IMAP evaluation and EO-based matched comparison group sampling



Salisbury et al. 2024 – earthquake and other disaster relief; conflict monitoring in Ukraine



Multi-decadal availability of Landsat imagery

- Planet’s SkySat satellites can revisit up to 10x/day

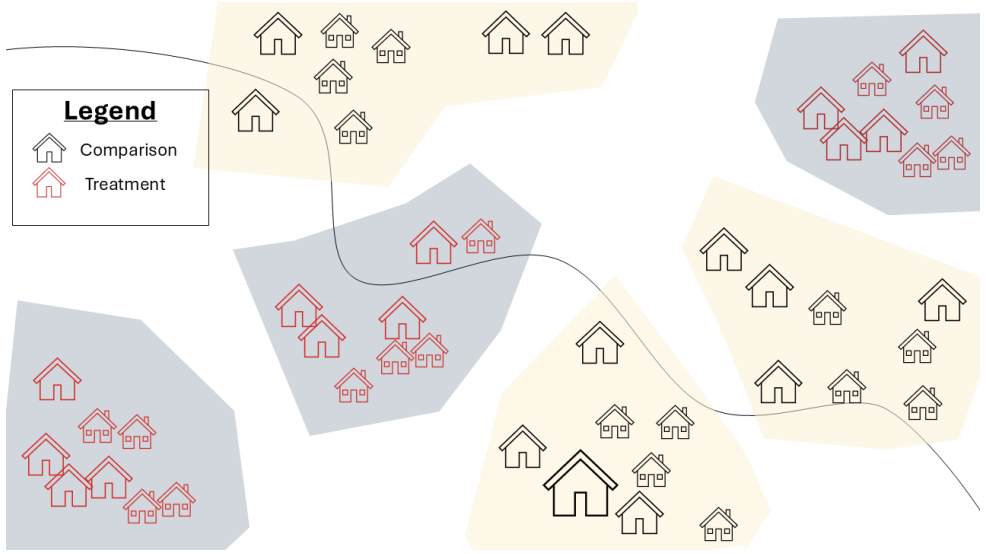
Randomized controlled trial (RCT) | Balance on observables and unobservables

- Social science method most closely approximating a scientific "experiment" and considered the methodological "gold standard", but can't always be carried out
  - Ethical issues; feasibility; cost; timeframe
- Ex-ante methodology → conduct power calculations to determine sample size and minimum detectible effect
- Can verify success of randomization through baseline balance tests, if baseline data is collected

Individual randomization

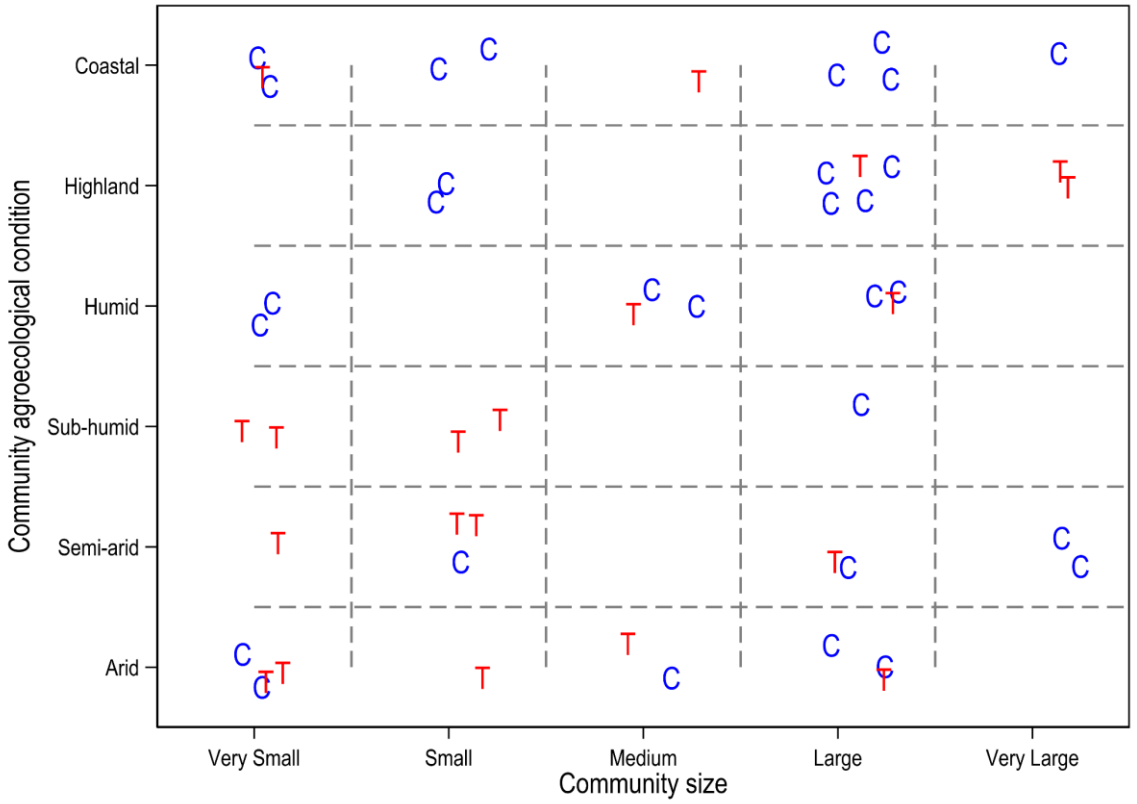


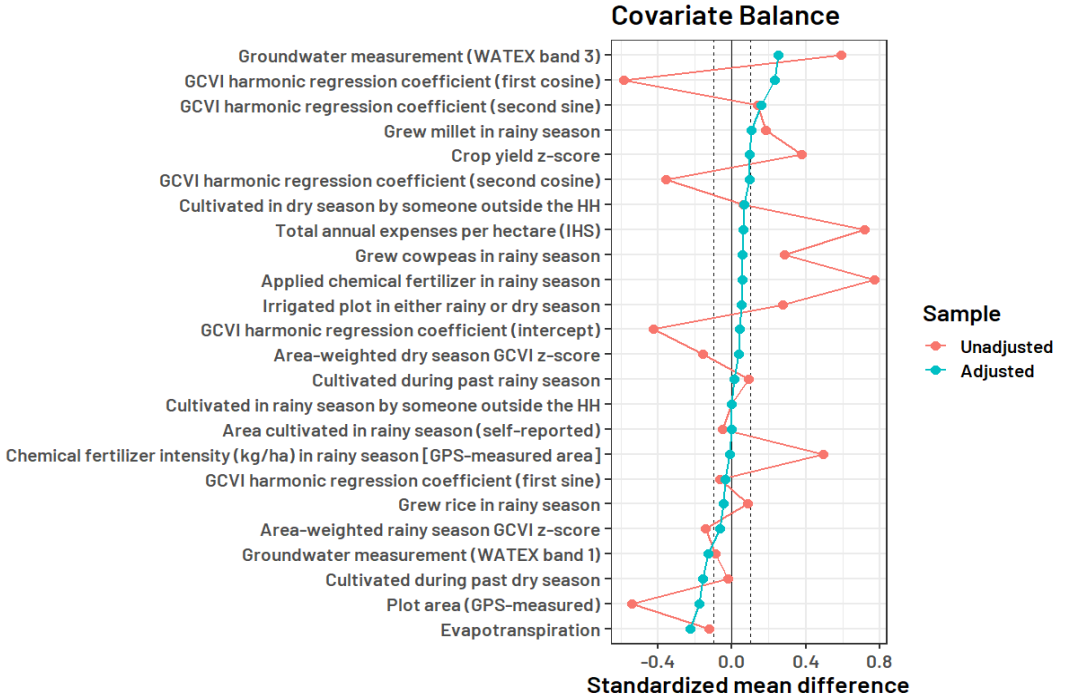
Cluster-level randomization

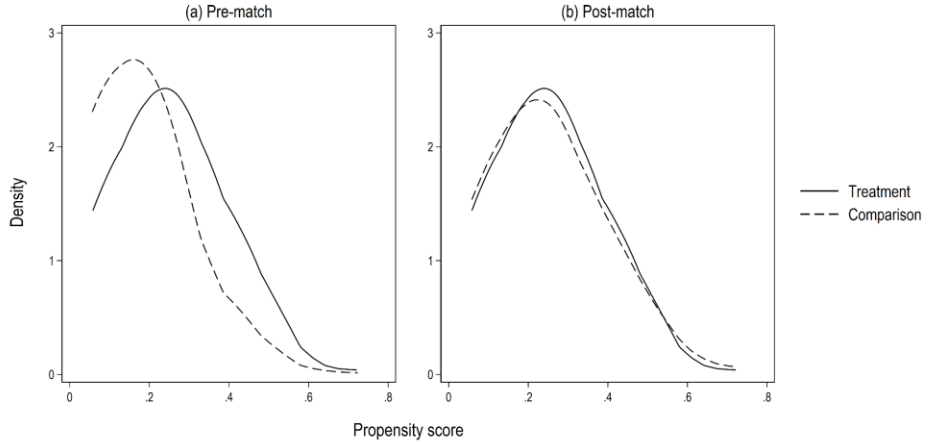


Matching methods | Selecting a comparison group and conducting balance tests

- Though an ex-post design, researchers should make comparison selection decisions based only on pre-intervention data



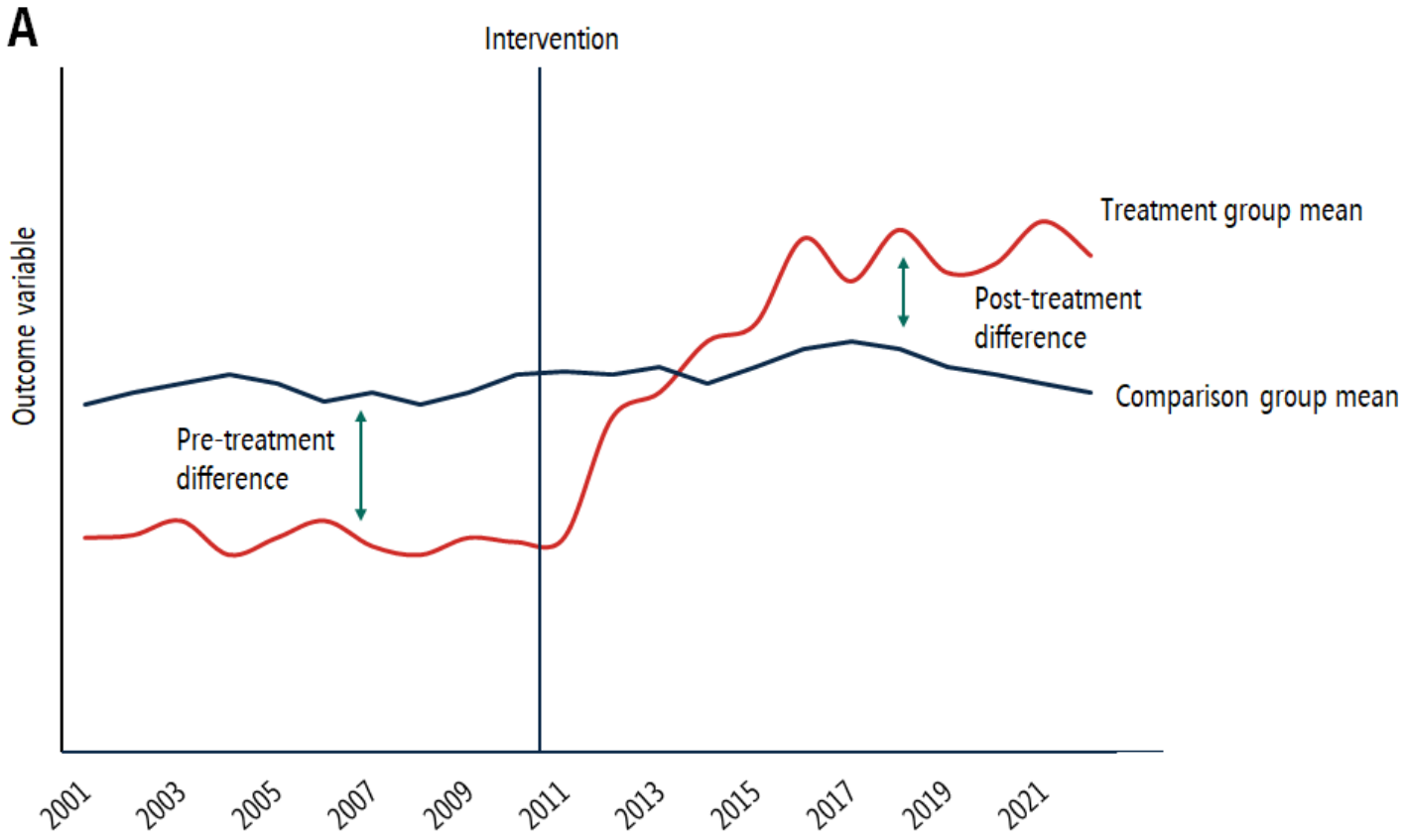


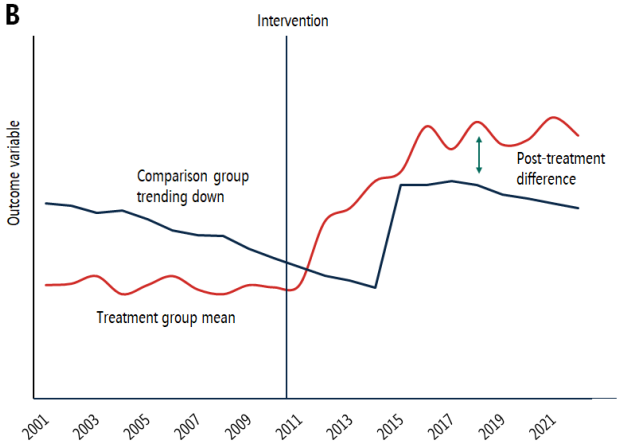


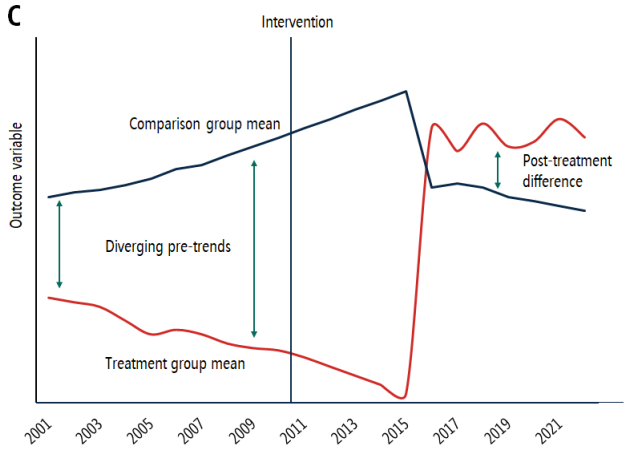
Mathematica
Progress Together

Difference in differences (DID) | Parallel pre-trends and example violations

- **Goal**: Help people learning DID for the first time understand possible temporal dynamics

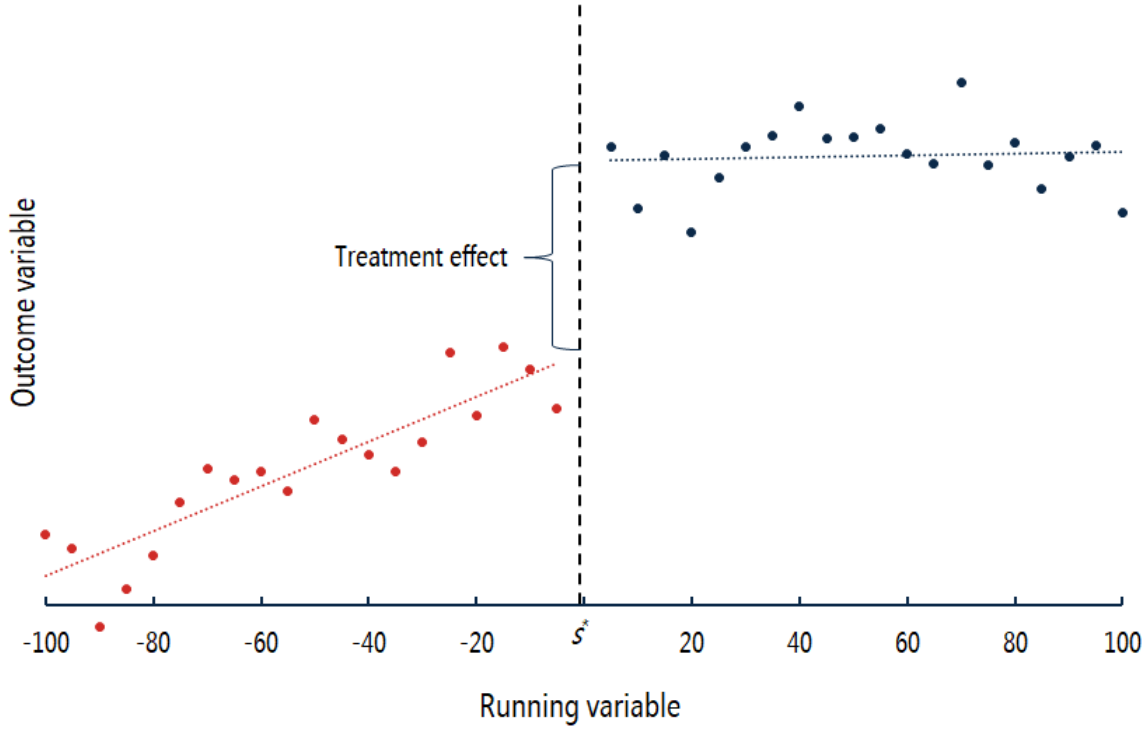




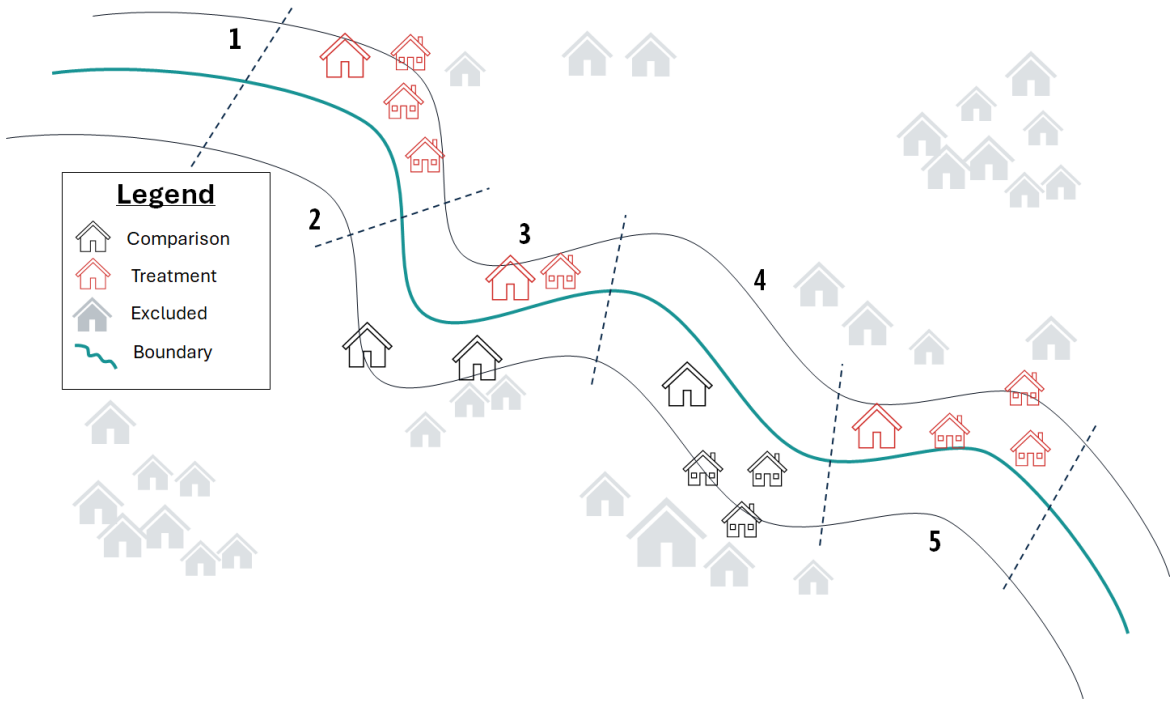


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Regression discontinuity (RD) | Extension to a spatial RD setup

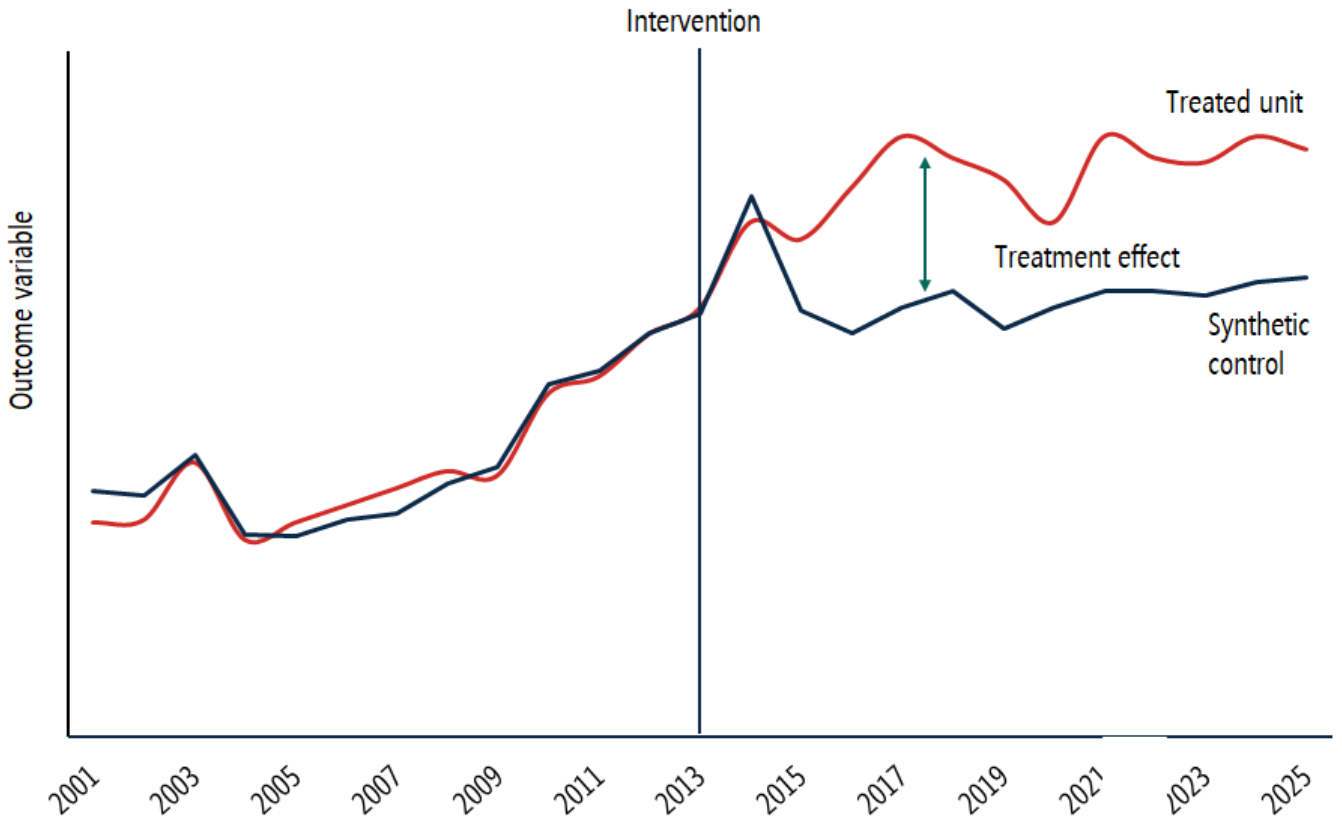


- 1D RD cutoff – discontinuous kink at threshold value, not known to units



- 2D SRD cutoff – T+C units restricted to defined spatial boundary
- Apply piece-wise approach, estimating distinct treatment effects in 2D space across neighborhoods

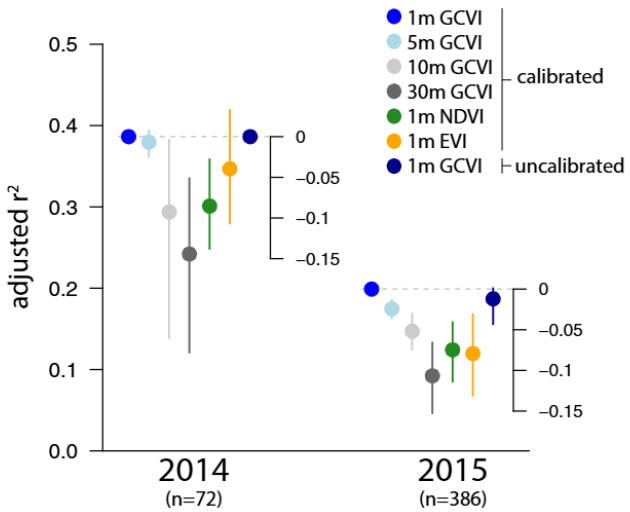
Synthetic control method (SCM) | Constructing a synthetic control from a donor pool



- Quant "case study" approach: can be $N_T = 1$
- Specify the untreated units that comprise the donor pool
- Weights are algorithmically selected to minimize a loss function comparing *k* predictor variables in pre-intervention period
- Visual inspection clarifies whether where the synthetic comparison is "observationally equivalent" to the treated unit in pre-intervention period
- Serra-Burriel et al. 2021 (*RSE*) – exclusively uses EO data in analyzing wildfire effects on vegetation recovery in California

Measurement and data considerations | Generating robust EO results

Recognizing proxies ≠ outcomes (always)



Testing different VIs on crop yields

Burke and Lobell 2017 (PNAS)

When admin data is manipulated or incomplete



USA animal farming operations (AFOs)

Palandri et al. *forthcoming*



Bangladesh brick kilns

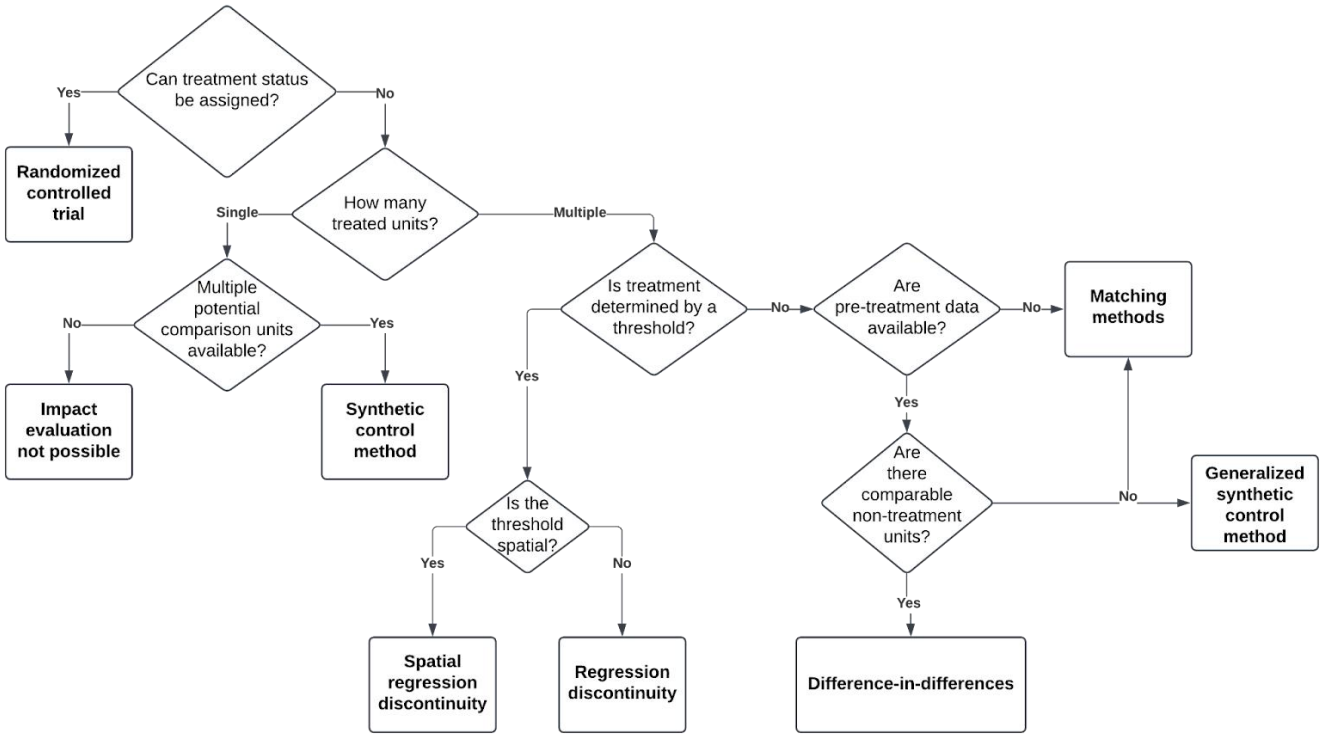
Lee et al. 2021 (PNAS)

Spillover effects and SUTVA violations



Biases treatment effect estimates because comparison group receives some indirect treatment; violates treatment uniformity assumption

Decision tree and checklist to highlight most suitable IE method



1. **Treatment definition and timing**
   a. Is the treatment clearly defined and observable?
   b. Can the timing of treatment be precisely determined?
   c. Is treatment assignment simultaneous or staggered across units?
   d. Is treatment binary or continuous?
2. **Comparison group construction**
   a. Are there clear potential comparison units?
   b. Are comparison units likely to be affected by spillovers from treatment?
   c. What is the appropriate geographic scale for comparison units?
3. **Data availability**
   a. What is the temporal coverage relative to treatment timing?
   b. What is the spatial resolution relative to the unit of treatment?
   c. Are there gaps in coverage (e.g., due to clouds or other factors)?
   d. Are complementary data sources available for validation?
4. **Measurement quality**
   a. How well does the EO measure proxy for the outcome of interest?
   b. Is the measurement approach stable across time and space?
   c. Are there known sources of measurement error?
   d. Can the measurement approach be validated?

Exercise questions | Have we adequately explained the key concepts?

1. Why is the randomized controlled trial considered the “gold standard” methodology?
2. What are two advantages that EO data offer for conducting impact evaluations beyond traditional data sources? What are key limitations or factors about EO data that researchers should be aware of being using it in their GIE?
3. If treatment assignment has not yet happened for an intervention, which research methodology would you advocate for? Are there any reasons this methodology might be inappropriate?
4. How is the average treatment effect of the DID method calculated?
5. How would you determine whether an SCM method is appropriate for a given use case?
6. What is the conceptual problem that matching methods aim to solve?
7. What is the purpose of having a comparison group? How would one determine whether the comparison group is serving its purpose?
8. In an RD framework, what determines which units are included in the analysis?



Thank you for your comments, suggestions, and questions!

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